NUMERICAL ANALYSIS
PROBLEM SET 3

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We begin by noting that all of the files referenced in this document are available at:
https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet3

1. Problem 1

We'll be approximating $\sqrt{2}$ though this problem

1.a. Our goal is to find a polynomial interpolant $p(x)$ using the Lagrange form given that we have three points in the form $x, f(x)$ as $\{(-1, 1/2), (0, 1), (1, 2)\}$. So we let $f(x) = 2^x$ such that $f(1/2) = \sqrt{2}$. So our polynomial begins as

$$p(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x),$$

with $y_i = f(x_i)$ and $L_i(x)$ being the i^{th} Lagrange polynomial. Computing the polynomials follows as

$$L_{2,0}(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 0)(x - 1)}{(-1 - 0)(-1 - 1)} = \frac{x(x - 1)}{2}$$
$$L_{2,1}(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x + 1)(x - 1)}{(0 + 1)(0 - 1)} = 1 - x^2$$
$$L_{2,2}(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x + 1)(x - 0)}{(1 + 1)(1 - 0)} = \frac{x(x + 1)}{2}$$

1.b. Putting these together we have the following polynomial:

$$\begin{aligned} p(x) &= \frac{1}{2} \cdot \frac{x(x - 1)}{2} + 1 \cdot (1 - x^2) + 2 \cdot \frac{x(x + 1)}{2} \\ &= \frac{x^2}{4} - \frac{x}{4} + 1 - x^2 + x^2 + x \\ &= \frac{x^2}{4} + \frac{3x}{4} + 1 \end{aligned}$$

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Pluggin in $x = 1/2$ we get the following

$$\begin{aligned} p\left(\frac{1}{2}\right) &= \frac{1}{4}\cdot\frac{1}{4}+\frac{3}{4}\cdot\frac{1}{2}+1 \\ &= \frac{1}{16}+\frac{3}{8}+1 \\ &= \frac{1}{16}+\frac{6}{16}+\frac{16}{16} \\ &= \frac{23}{16} \approx \sqrt{2}. \end{aligned}$$

1.c. For the error bound we want to compute the following

$$|f(x)-p(x)| = \frac{f'''(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2),$$

where $\xi \in [-1,1]$. We compute the third derivative of $f(x)$ by setting $f(x) = \ln(x) = e^{x\ln(2)}$. Clearly then we have

$$f'''(x) = (\ln(2))^3 e^{x\ln(2)} = (\ln(2))^3 2^x.$$

We want to have ξ such that $|f'''(\xi)|$ is largest so that everything else is beneath that as our upper bound. Because $f(x) = 2^x$ is an increasing exponential, the maximum of the third derivative will necessarily be the right hand end point. So we set $\xi = 1$ giving us $f'''(\xi) = 2(\ln(2))^3$. So

$$|f(x)-p(x)| = \frac{f'''(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2) = \frac{2(\ln(2))^3}{6}(x+1)(x)(x-1)$$

And we want to know the error at $x = 1/2$ which comes out at

$$\begin{aligned} \left|f\left(\frac{1}{2}\right)-p\left(\frac{1}{2}\right)\right| &= \frac{2(\ln(2))^3}{6}\left(\frac{1}{2}+1\right)\cdot\frac{1}{2}\cdot\left(\frac{1}{2}-1\right) \\ &= \frac{3\ln(2)^3}{24} \\ &= \frac{\ln(2)^3}{8} = 0.0416 \end{aligned}$$